Table 2: Node-level comparison with representative asynchronous NoC routers.

Design	**Technology**	**Flit Width**	**Latency (ns)**	**Throughput[a](Gb/s)**	**Active Power[b](mW)**	**Area (μm^2)**
CMR [1]	45 nm	128	0.693[c]	–	–	54,083
TaBuLA [2]	40 nm	32	0.980[d]	34.24	–	24,866
Zeng *et al.* [3]	55 nm	4	1.250[d]	4.00	–	22,400
Reusable [4]	22 nm	4	0.560[c]	–	–	9,902
This work (Thin, Async c1p1)	**28 nm**	**28**	**0.710**[c]	**33.53**	**1.578**	**7,293**
This work (Async c1p4)	**28 nm**	**28**	**1.101**[c]	**84.04**	**1.702**	**12,602**
This work (Sync c1p4)	**28 nm**	**28**	**1.000**[c]	**–**	**7.167**	**11,374**

[a] Logical data throughput. Values are converted from the reported flit rate (or equivalent speed) using the corresponding flit width. For this work, Async c1p1 and c1p4 achieve 1.197 and 3.001 Gflit/s, corresponding to 33.53 and 84.04 Gb/s, respectively.

[b] Absolute active power is reported only where directly available. Values for this work are obtained from time-based PrimeTime PX. "–" indicates that a directly comparable absolute power value is not reported.

[c] Head-flit / head-hop latency.

[d] Reported average forward/flit latency. Latency definitions vary slightly across prior works; values are therefore shown as reported rather than technology-normalized.